Ejemplos

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Bank notes

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.5.1
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```r
library(webshot2)

data(banknote, package = "mclust")
swissTib <- as_tibble(banknote)

head(swissTib)
```

```
## # A tibble: 6 x 7
##   Status  Length  Left Right Bottom   Top Diagonal
##   <fct>    <dbl> <dbl> <dbl>  <dbl> <dbl>    <dbl>
## 1 genuine   215.  131   131.    9     9.7     141
## 2 genuine   215.  130.  130.    8.1   9.5     142.
## 3 genuine   215.  130.  130.    8.7   9.6     142.
## 4 genuine   215.  130.  130.    7.5  10.4     142
## 5 genuine   215   130.  130.   10.4   7.7     142.
## 6 genuine   216.  131.  130.    9    10.1     141.
```

```r
# La matriz de covarianza
cov(swissTib[, 2:7])
```

```
##               Length        Left       Right     Bottom        Top    Diagonal
## Length    0.14179296  0.03144322  0.02309146 -0.1032462 -0.0185407  0.08430553
## Left      0.03144322  0.13033945  0.10842739  0.2158028  0.1050394 -0.20934196
## Right     0.02309146  0.10842739  0.16327412  0.2841319  0.1299967 -0.24047010
## Bottom   -0.10324623  0.21580276  0.28413191  2.0868781  0.1645389 -1.03699623
```

```
## Top      -0.01854070  0.10503945  0.12999673  0.1645389  0.6447234 -0.54961482
## Diagonal  0.08430553 -0.20934196 -0.24047010 -1.0369962 -0.5496148  1.32771633
```

```r
# No es necesariamente la que pca usa, pues hace falta centrar y escalar los datos

# Quitamos la columna de etiquetas
pca <- select(swissTib, -Status) %>%
  prcomp(center = TRUE, scale = TRUE)
# Los datos estan centrados y escalados (normalizados) [como normalizar un vector]
```

Las proyecciones están en:

```r
pca$x %>% head
```

```
##             PC1         PC2        PC3        PC4          PC5         PC6
## [1,] -1.7430272  1.64669605  1.4201973 -2.7479691  0.003293759  0.60202200
## [2,]  2.2686248 -0.53744461  0.5313151 -0.6573558 -0.158171742  0.45654268
## [3,]  2.2717009 -0.10740754  0.7156191 -0.3408384 -0.453880889 -0.04532905
## [4,]  2.2778385 -0.08743490 -0.6041176 -0.3918255 -0.282913485 -0.05543875
## [5,]  2.6255397  0.03909779  3.1883837  0.4240168 -0.277502895  0.72026433
## [6,] -0.7565089  3.08101359  0.7845117 -0.5980322  0.192757017 -0.10529393
```

Los vectores propios están en (matriz U):

```r
pca$rotation
```

```
##                   PC1         PC2         PC3        PC4        PC5         PC6
## Length    0.006987029  0.81549497 -0.01768066  0.5746173 -0.0587961  0.03105698
## Left     -0.467758161  0.34196711  0.10338286 -0.3949225  0.6394961 -0.29774768
## Right    -0.486678705  0.25245860  0.12347472 -0.4302783 -0.6140972  0.34915294
## Bottom   -0.406758327 -0.26622878  0.58353831  0.4036735 -0.2154756 -0.46235361
## Top      -0.367891118 -0.09148667 -0.78757147  0.1102267 -0.2198494 -0.41896754
## Diagonal  0.493458317  0.27394074  0.11387536 -0.3919305 -0.3401601 -0.63179849
```

Podemos ver la desviación estándar en cada dirección principal con:

```r
pca$sdev
```

```
## [1] 1.7162629 1.1305237 0.9322192 0.6706480 0.5183405 0.4346031
```

Para obtener los valores propios (matriz Λ):

```r
(pca$sdev)^2
```

```
## [1] 2.9455582 1.2780838 0.8690326 0.4497687 0.2686769 0.1888799
```

La desviación estándar acumulada con:

```r
summary(pca)
```

```
## Importance of components:
##                           PC1    PC2    PC3     PC4     PC5     PC6
## Standard deviation     1.7163 1.1305 0.9322 0.67065 0.51834 0.43460
## Proportion of Variance 0.4909 0.2130 0.1448 0.07496 0.04478 0.03148
## Cumulative Proportion  0.4909 0.7039 0.8488 0.92374 0.96852 1.00000
```

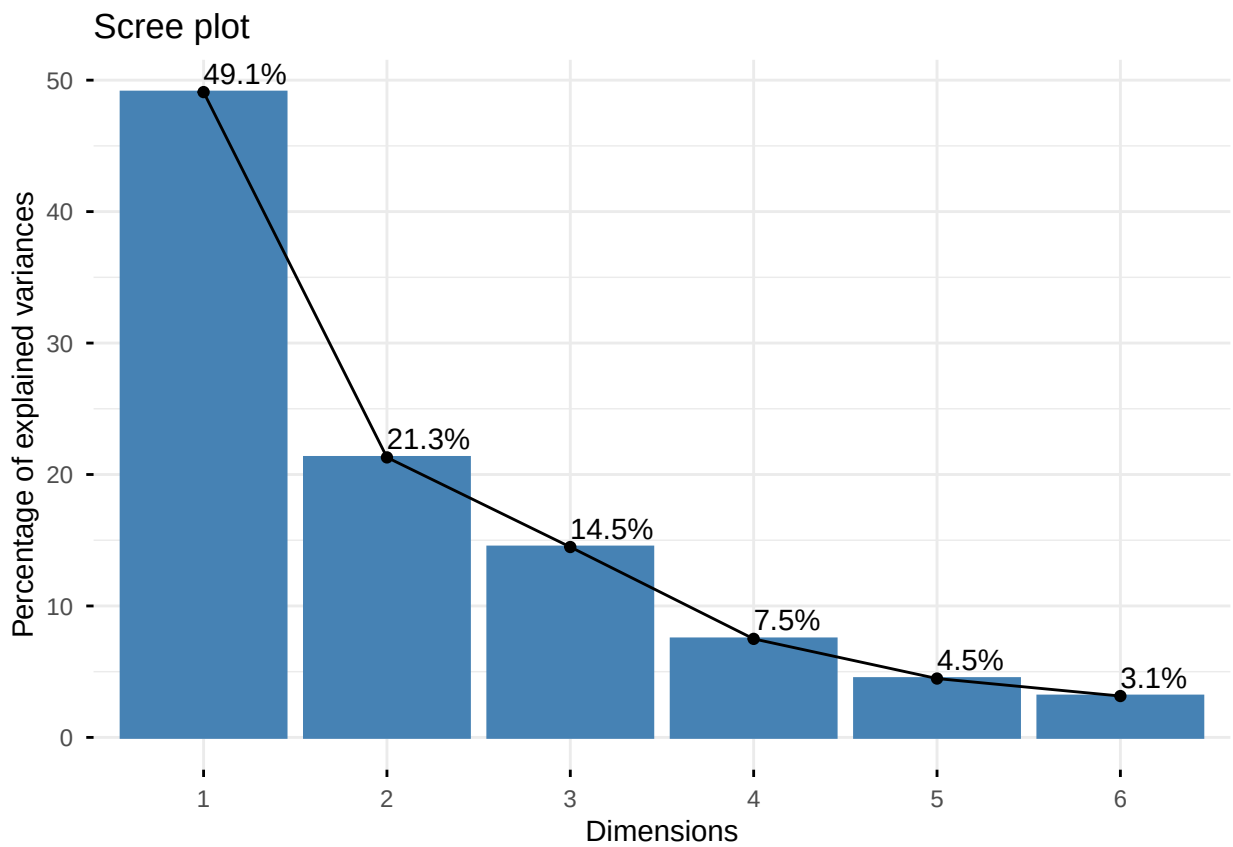
Graficamos las primeras dos componentes principales:

```r
library(plotly)
```

```
##
## Attaching package: 'plotly'
```

```
## The following object is masked from 'package:ggplot2':
##
##     last_plot

## The following object is masked from 'package:stats':
##
##     filter

## The following object is masked from 'package:graphics':
##
##     layout
```

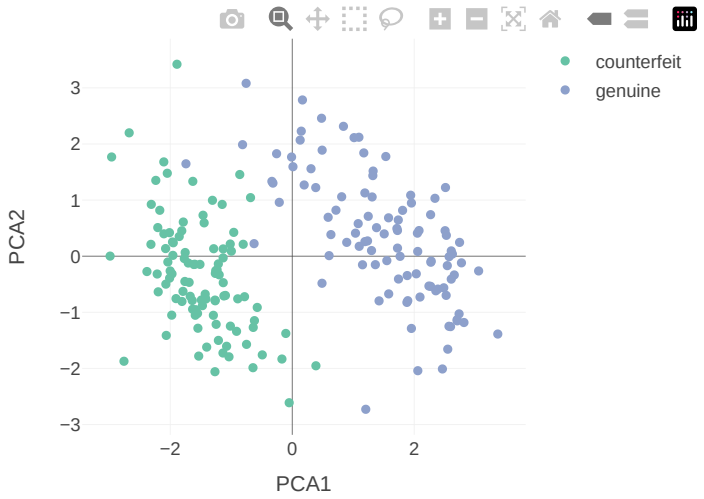
```
fpca <- get_pca(pca)

fviz_screeplot(pca, choice="variance", addlabels=TRUE)
```




```
swiss2Pca <- swissTib %>%
  mutate(PCA1 = pca$x[, 1], PCA2 = pca$x[, 2])

plot_ly(data=swiss2Pca, x=~PCA1, y=~PCA2, color=~Status)
```

```
## No trace type specified:
##   Based on info supplied, a 'scatter' trace seems appropriate.
##   Read more about this trace type -> https://plotly.com/r/reference/#scatter

## No scatter mode specifed:
##   Setting the mode to markers
##   Read more about this attribute -> https://plotly.com/r/reference/#scatter-mode
```

```
## file:////tmp/RtmpBcxbzk/file181d4514263e/widget181d22269654.html screenshot completed
```



Graficamos tres componentes principales:

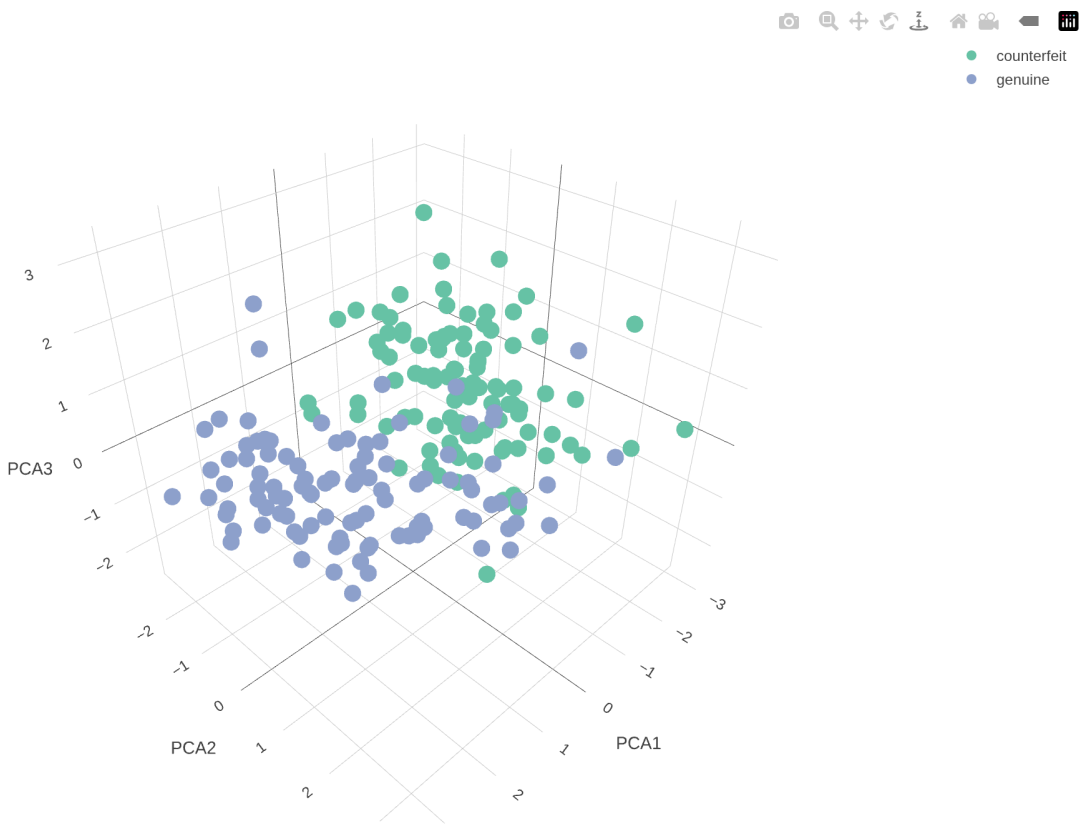
```r
library(htmlwidgets)
library(plotly)
library(webshot2)
Sys.setenv(CHROMOTE_CHROME = "/usr/bin/google-chrome")

swiss3PCA <- swissTib %>%
  mutate(PCA1 = pca$x[, 1], PCA2 = pca$x[, 2], PCA3 = pca$x[, 3])
p <- plot_ly(data=swiss3PCA, x=~PCA1, y=~PCA2, z=~PCA3, color=~Status)

# 2. Save the plotly object as a standalone HTML file
temp_html <- "temp_plot.html"
saveWidget(p, temp_html, selfcontained = TRUE)
```

```
## No trace type specified:
##   Based on info supplied, a 'scatter3d' trace seems appropriate.
##   Read more about this trace type -> https://plotly.com/r/reference/#scatter3d
```

```
## No scatter3d mode specifed:
##   Setting the mode to markers
##   Read more about this attribute -> https://plotly.com/r/reference/#scatter-mode
```

```r
# 3. Use webshot2 to take the picture (where delay/zoom actually work!)
webshot2::webshot(
  url = temp_html,
  file = "3d_snapshot.png",
  delay = 3,    # Wait 3 seconds for WebGL to load
  zoom = 2,     # Double the resolution for the PDF
  vwidth = 1000,
  vheight = 700
)
```

```
## file:////home/akatl/Lambda/proyecto-terminal/code/temp_plot.html screenshot completed
```




```
# 4. Display the image in your PDF
#knitr::include_graphics("3d_snapshot.png")
```

Agregar nuevos datos

```
newBanknotes <- tibble(
  Length = c(214, 216),
  Left = c(130, 128),
  Right = c(132, 129),
  Bottom = c(12, 7),
  Top = c(12, 8),
  Diagonal = c(138, 142)
)
predict(pca, newBanknotes)
```

```
##            PC1        PC2       PC3       PC4       PC5      PC6
## [1,] -4.729496 -1.9989069 0.1058105 -1.658777 -3.202514 1.623096
## [2,]  6.465758  0.8918486 0.8214539  3.468704 -1.837978 2.339436
```